MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 21: Evaluation of Real Integrals

Nick Strigul
Washington State University Vancouver
nick.strigul@wsu.edu

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Lecture outline

1. Integrals of the Form $I = \int_0^{2\pi} R(\cos\theta, \sin\theta)d\theta$
2. Improper Integrals of Rational Functions
3. $\int_{-\infty}^{\infty} f(x)\cos axdx$ and $\int_{-\infty}^{\infty} f(x)\sin axdx$.

Integrals of the Form $I = \int_0^{2\pi} R(\cos\theta, \sin\theta)d\theta$. Part 1

The residue theorem can be used to evaluate certain kinds of real integrals. Let us consider integrals of the form

$$I = \int_0^{2\pi} R(\cos\theta, \sin\theta)d\theta$$

Here $R(s,t)$ is a rational function of s and t. We assume that $R(\cos\theta, \sin\theta)$ is finite on $0 \leq \theta \leq 2\pi$. also Let $z = e^{i\theta}$. Then the interval $0 \leq \theta \leq 2\pi$ yields the

curve $C : |z| = 1$.
Now

$$\cos\theta = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

and

$$\sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right) = \frac{1}{2i}\left(z - \frac{1}{z}\right)$$

Integrals of the Form $I = \int_0^{2\pi} R(\cos\theta, \sin\theta)d\theta$. Part 2.

Then

$$dz = ie^{i\theta}d\theta = izd\theta$$

or

$$d\theta = \left(\frac{1}{i}\right)\left(\frac{dz}{z}\right)$$

Thus

$$I = \oint_C R\left[\frac{1}{2}\left(z + \frac{1}{z}\right), \frac{1}{2i}\left(z - \frac{1}{z}\right)\right]\frac{1}{iz}dz$$

where $C : |z| = 1$.

The integral I may be evaluated by residues.

Example 1. Part 1.

Evaluate

$$\int_0^{2\pi} \frac{\cos 2\theta}{1 - 2p\cos\theta + p^2}\, d\theta \qquad -1 < p < 1$$

Note that by adding and subtracting $2p$, the denominator of the integrand can be written in either of two equivalent forms:

$$\begin{aligned} 1 - 2p\cos\theta + p^2 &= 1 - 2p + p^2 + 2p - 2p\cos\theta \\ &= (1-p)^2 + 2p\,(1 - \cos\theta) \end{aligned}$$

$$\begin{aligned} 1 - 2p\cos\theta + p^2 &== 1 + 2p + p^2 - 2p - 2p\cos\theta \\ &= (1+p)^2 - 2p\,(1 + \cos\theta) \end{aligned}$$

From the first of these it is clear that if $0 \le p < 1$, the denominator is different from zero for all values of θ; and from the second it is clear that if $-1 < p \le 0$, the denominator is also different from zero for all values of θ.

Example 1. Part 2.

Therefore the integral becomes:

$$\oint_C \frac{z^2+z^{-2}}{2} \frac{1}{1-\frac{2p\left(z+z^{-1}\right)}{2}+p^2} \frac{dz}{iz} =$$

$$= \oint \frac{z^4+1}{2z^2} \frac{z}{z-pz^2-p+p^2z} \frac{dz}{iz}$$

$$= \oint_C \frac{\left(1+z^4\right) dz}{2iz^2\left(1-pz\right)\left(z-p\right)}$$

$$= \oint_C f\left(z\right) dz$$

There are three poles in the integrand at 0, $\frac{1}{p}$, and p. Now $-1 < p < 1$ so that $\frac{1}{p}$ is outside of $|z| = 1$. Therefore we may disregard this pole.

Example 1. Part 3.

For the other two poles we have:

$$\mathrm{Res}(f,p) = \lim_{z\to p}(z-p)\,\frac{\left(1+z^4\right)}{2iz^2\left(1-pz\right)\left(z-p\right)} = \frac{1+p^4}{2ip^2\left(1-p^2\right)}$$

The pole at 0 is second order and therefore

$$\mathrm{Res}(f,0) = \lim_{z\to 0}\frac{d}{dz}\left[z^2\frac{\left(1+z^4\right)}{2iz^2\left(1-pz\right)\left(z-p\right)}\right]$$

$$= \lim_{z\to 0}\frac{\left(z-pz^2-p+p^2z\right)\left(4z^3\right)-\left(1+z^4\right)\left(1-2pz+p^2\right)}{2i\left(z-pz^2-p+p^2z\right)^2} = -\frac{1+p^2}{2ip^2}$$

Thus $\quad I = 2\pi i\left[\frac{1+p^4}{2ip^2\left(1-p^2\right)}-\frac{1+p^2}{2ip^2}\right] = \frac{2\pi p^2}{1-p^2}$

Improper Integrals of Rational Functions. Part 1.

Consider

$$\int_{-\infty}^{\infty} f(x)\,dx$$

where

$$f(x) = \frac{P(x)}{Q(x)}$$

is a rational function and the

$$\deg Q(x) \geq \deg P(x) + 2.$$

Furthermore, suppose $Q(x)$ has no real zeros. Such an integral is called an improper integral since the interval of integration is not finite. We define

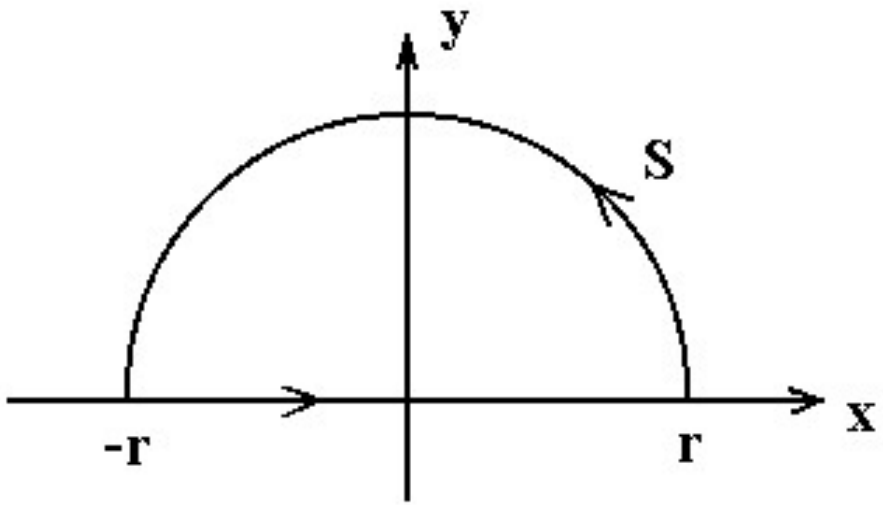
$$\int_{-\infty}^{\infty} f(x)\,dx = \lim_{r\to\infty} \int_{-r}^{r} f(x)\,dx$$

Improper Integrals of Rational Functions. Part 2.

To evaluate this integral consider the contour integral

$$\oint_C f(z)\,dz$$

where C is the contour below.



Since $f(x)$ is a rational function, this implies that $f(z) = \frac{P(z)}{Q(z)}$ has finitely many poles in the upper half plane. These are the zeros of $Q(z)$, say $z_1, \ldots, z_n$. We choose r large enough so that C encloses all of these zeros.

Improper Integrals of Rational Functions. Part 3.

Then

$$\oint_C f(z)\,dz = \int_S f(z)\,dz + \int_{-r}^{r} f(x)\,dx = 2\pi i \sum_{j=1}^{n} \operatorname{Res}(f, z_j).$$

Therefore

$$\int_{-r}^{r} f(x)\,dx = 2\pi i \sum_{j=1}^{n} \operatorname{Res}(f, z_j) - \int_S f(z)\,dz$$

.

It turns out that

$$\int_S f(z)\,dz \to 0$$

as $r \to \infty$. On S, if we let $z = re^{i\theta}$, then S is represented by $r = const$, $0 \leq \theta\pi$.

Improper Integrals of Rational Functions. Part 4.

Since

$$\deg Q(x) \geq \deg P(x) + 2,$$

then

$$|f(z)| < \frac{k}{|z|^2} \qquad \text{for } |z| = r > r_0,$$

where r_0 and k are sufficiently large. Hence

$$\left| \int_S f(z)\, dz \right| < \frac{k}{r^2}(\pi r) = \frac{k\pi}{r} \qquad \text{for } r > r_0$$

and this $\to 0$ as $r \to \infty$, so that

$$\int_{-\infty}^{\infty} f(x)\, dx = 2\pi i \sum_{j=1}^{n} \operatorname{Res}(f, z_j)$$

Example 2.

Evaluate

$$\int_{-\infty}^{\infty} \frac{x^2}{\left(x^2+a^2\right)\left(x^2+b^2\right)} dx$$

where $a, b > 0$.

The poles of

$$\frac{x^2}{\left(x^2+a^2\right)\left(x^2+b^2\right)}$$

are at $z = \pm ai, \pm bi$. Of these only $z = ai$ and $z = bi$ lie in the upper half plane. At $z = ai$ the residue is:

$$\lim_{z \to ai} (z - ai) \frac{z^2}{(z-ai)(z+ai)\left(z^2+b^2\right)} = \frac{-a^2}{2ai\left(-a^2+b^2\right)} = \frac{a}{2i\left(a^2-b^2\right)}$$

Similarly, at $z = bi$ we have $\frac{b}{2i(b^2-a^2)}$. Hence the value of the integral is

$$2\pi i \left[\frac{a}{2i\left(a^2-b^2\right)} - \frac{b}{2i\left(a^2-b^2\right)}\right] = \frac{\pi}{a+b}.$$

$\int_{-\infty}^{\infty} f(x)\cos axdx$ and $\int_{-\infty}^{\infty} f(x)\sin axdx$. Part 1.

Here $f(x)$ is a rational functional with the properties listed above.
We note that

$$\int_{-\infty}^{\infty} f(x)\cos axdx + i\int_{-\infty}^{\infty} f(x)\sin axdx = \int_{-\infty}^{\infty} f(x)\, e^{iax}dx$$

The

$$\int_{-\infty}^{\infty} f(x)\, e^{iax}dx$$

may be treated as above. Thus

$$\int_{-\infty}^{\infty} f(x)\, e^{iax}dx = 2\pi i\sum \text{residues of } e^{iaz}f(z) \text{ at its poles in the upper half plane}$$

$\int_{-\infty}^{\infty} f(x)\cos axdx$ and $\int_{-\infty}^{\infty} f(x)\sin axdx$. Part 2.

Therefore

$$\int_{-\infty}^{\infty} f(x)\cos ax\, dx =$$

$$= -2\pi \sum \text{imaginary parts of residues of } e^{iaz} f(z) \text{ at its poles in the upper half plane}$$

Similarly

$$\int_{-\infty}^{\infty} f(x)\sin ax\, dx =$$

$$= 2\pi \sum \text{real parts of the residues of } e^{iaz} f(z) \text{ at its poles in the upper half plane}$$

Example 3.

Show that for $a > 0, k > 0$

$$\int_{-\infty}^{\infty} \frac{\cos ax}{k^2 + x^2} dx = \frac{\pi}{k} e^{-ak}$$

and

$$\int_{-\infty}^{\infty} \frac{\sin ax}{k^2 + x^2} dx = 0.$$

In this case

$$f(z) = \frac{e^{iaz}}{k^2 + z^2}$$

has only one pole in the upper half plane, namely a single pole at $z = ik$, then

$$\mathrm{Res}(f(z), ik) = \lim_{z \to ik} (z - ik) \frac{e^{iaz}}{(z - ik)(z + ik)} = \frac{e^{-ka}}{2ik}.$$

and the following yields the result:

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{k^2 + x^2} dx = 2\pi i \frac{e^{-ka}}{2ik} = \frac{\pi}{k} e^{-ka}$$

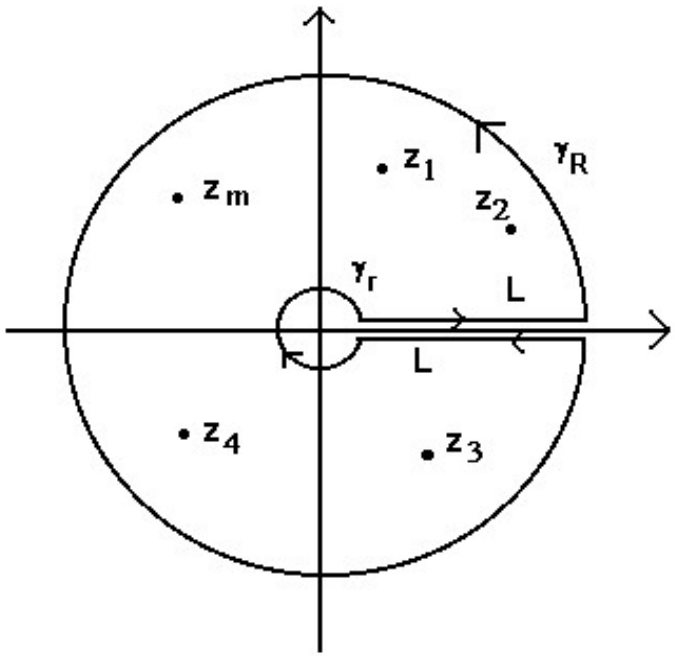
Integrals $\int_0^\infty x^a p(x)/q(x)\,dx$. Part 1.

Let a be a real number with $0 < a < 1$ and consider integrals of the form

$$\int_0^\infty \frac{x^a p(x)}{q(x)}\,dx,$$

in which p and q are polynomials with real coefficients and no common factors. Furthermore, q has no real positive zeros, the degree of q exceeds the degree of p by at least 1, and either $q(0) \neq 0$ or q has a simple zero at 0.

Integrals $\int_0^\infty x^a p(x)/q(x)\,dx$. Part 2.



Let the nonzero zeros of q be $z_1, \ldots, z_m$. (Note that these are *all* the nonzero zeros of q, not just the zeros in the upper half-plane.) Choose r small enough and R large enough that the closed path Γ in the figure below encloses $z_1, \ldots, z_m$. The closed path Γ consists of γ_R, most of the circle of radius R about the origin; γ_r, part of the circle of radius r about the origin; and the line segments L_1 and L_2 connecting γ_R and γ_r. Eventually we shall let $R \to \infty$ as $r \to 0$.

Integrals $\int_0^\infty x^a p(x)/q(x)\,dx$. Part 3.

We must decide what we mean by z^a. Recall that this symbol generally denotes a (possibly infinite) set of different numbers. We write $z = \rho e^{i\theta}$ for $0 \le \theta < 2\pi$ so

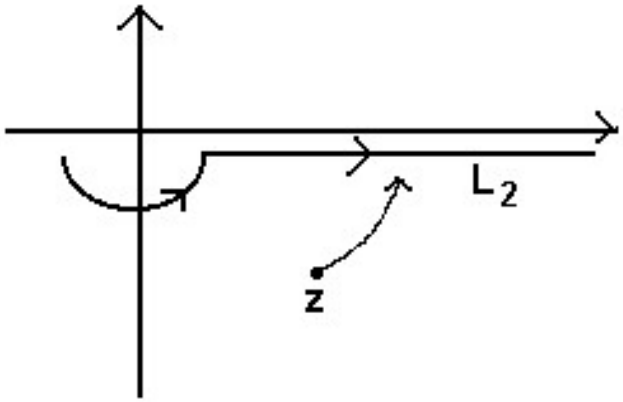
$$z^a = \rho^a e^{ia\theta}.$$

As z approaches L_1,

$$f(z) = \frac{z^a p(z)}{q(z)} \to \frac{x^a p(x)}{q(x)},$$

where $r < x < R$. However, as z approaches L_2 (the lower side of the positive real axis) as in the figure below, then

$$f(z) \to \frac{x^a e^{2\pi a i} p(x)}{q(x)}.$$



Integrals $\int_0^\infty x^a p(x)/q(x)\,dx$. Part 4.

The reason for this is that the argument increases by 2π as z approaches the positive real axis from below, and then

$$z^a = \rho^a e^{i(\theta+2\pi)a} = \rho^a e^{ia\theta} e^{2\pi a i}.$$

By the residue theorem

$$\oint_\Gamma \frac{z^a p(z)}{q(z)} dz = 2\pi i \sum_{j=1}^{m} \operatorname{Res}(f, z_j)$$

$$= \int_{\gamma_R} \frac{z^a p(z)}{q(z)} dz + \int_{\gamma_r} \frac{z^a p(z)}{q(z)} dz + \int_{L_1} \frac{x^a p(x)}{q(x)} dx + \int_{L_2} \frac{x^a e^{2\pi a i} p(x)}{q(x)} dx.$$

On L_1 x varies from r to R, while on L_2 x varies from R to r to maintain the counterclockwise orientation of the contour.

Integrals $\int_0^\infty x^a p(x)/q(x)\,dx$. Part 5.

Now

$$2\pi i \sum_{j=1}^{m} \operatorname{Res}(f, z_j) = \int_{\gamma_R} \frac{z^a p(z)}{q(z)} dz + \int_{\gamma_r} \frac{z^a p(z)}{q(z)} dz + \\ + \int_r^R \frac{x^a p(x)}{q(x)} dx + \int_R^r \frac{x^a e^{2\pi a i} p(x)}{q(x)} dx.$$

By making estimates on the two circular arcs one can show that the two integrals in the above go to zero as $r \to 0$ and $R \to \infty$. In this limit the last equation yields

$$2\pi i \sum_{j=1}^{m} \operatorname{Res}(f, z_j) = + \int_0^\infty \frac{x^a p(x)}{q(x)} dx - \int_0^\infty \frac{x^a e^{2\pi a i} p(x)}{q(x)} dx.$$

Hence we finally have

$$\int_0^\infty \frac{x^a p(x)}{q(x)} dx = \frac{2\pi i}{1 - e^{2\pi a i}} \sum_{j=1}^{m} \operatorname{Res}(f, z_j).$$

Keep in mind that this formula is valid only when $0 < a < 1$, $z_1, \ldots, z_m$ are all the nonzero zeros of q, and that either $q(0) \neq 0$ or q has a simple zero at 0.

Example 4. Part 1.

Show that

$$\int_0^\infty \frac{x^{1/3}}{x\left(x^2+1\right)}dx = \pi$$

Here $q(z) = z\left(z^2+1\right)$ has simple zeros at $0, i$, and $-i$. Let

$$f(z) = \frac{z^{1/3}}{z\left(z^2+1\right)} = \frac{h(z)}{q(z)}.$$

Then

$$\mathrm{Res}(f,i) = \frac{h(i)}{q'(i)} = \frac{i^{1/3}}{3i^2+1} = \frac{\left(e^{\pi i/2}\right)^{1/3}}{-2} = -\frac{1}{2}e^{\pi i/6}$$

and

$$\mathrm{Res}(f,i) = \frac{h(i)}{q'(i)} = \frac{i^{1/3}}{3i^2+1} = \frac{\left(e^{\pi i/2}\right)^{1/3}}{-2} = -\frac{1}{2}e^{\pi i/6}$$

$$\mathrm{Res}(f,-i) = \frac{h(-i)}{q'(-i)} = \frac{i^{1/3}}{3(-i)^2+1} = \frac{\left(e^{3\pi i/2}\right)^{1/3}}{-2} = -\frac{1}{2}e^{\pi i/2}$$

Example 4. Part 2.

Hence

$$\int_0^\infty \frac{x^{1/3}}{x\left(x^2+1\right)}dx = \frac{2\pi i}{1-e^{2\pi a i}}\left(-\frac{1}{2}\right)\left(e^{\pi i/6}+e^{\pi i/2}\right)$$

$$= \frac{-\pi i}{1+1/2-\left(\sqrt{3}/2\right)i}\left(\frac{\sqrt{3}}{2}+\frac{1}{2}i+i\right) = \pi$$

Thank you